

EXPERIMENT NO. 1

Problem Statement & Software Process Model Justification

Project: Hospital Management System (HMS)

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1. Aim

To study the manual hospital administrative workflow, prepare a detailed Problem Statement for a computerized Hospital Management System (HMS), and select and justify a suitable software development process model for the project.

2. Objectives

- To study existing manual healthcare workflows and identify administrative bottlenecks and data integrity challenges.
- To define the scope, boundaries, and objectives of the proposed computerized Hospital Management System.
- To identify key stakeholders and define user classes (Patient, Doctor, Receptionist, Pharmacist, Lab Tech, Administrator).
- To evaluate and select an appropriate software development process model (V-Model / Agile Scrum) with comprehensive technical justification.

3. Problem Statement

In a traditional manual hospital setup, patient registration, doctor appointments, paper-based prescriptions, pharmacy inventory, laboratory test requisitions, and billing are maintained across independent physical ledgers and disjointed departmental records.

Key challenges in the manual system include:

1. Excessive Patient Wait Times: Manual registration and doctor scheduling create long queues at front desks and lead to double-booking collisions.
2. Data Redundancy & Lost Records: Paper charts are frequently misplaced, damaged, or unavailable across departments during emergencies.
3. Pharmacy Stock Discrepancies: Manual prescription dispensation leads to medication inventory mismatches, delayed reordering, and expired drug waste.
4. Billing Inaccuracies & Revenue Leakage: Disconnected billing ledgers between laboratory tests, pharmacy disbursements, and doctor consultation fees cause significant billing errors and delays during patient discharge.

5. Absence of Centralized Auditing: Hospital administrators lack real-time visibility into bed occupancy, daily patient footfall, and financial revenue.

4. Proposed Solution: Hospital Management System (HMS)

The proposed web-based Hospital Management System (HMS) provides a centralized, multi-tier database-driven platform connecting all clinical and administrative departments in real time.

Key modules include:

- Patient Registration & Appointment Scheduling (Token management, doctor calendar synchronization).
- Doctor Consultation & Electronic Health Records (Clinical notes, diagnosis history, electronic prescriptions).
- Diagnostic Laboratory Management (Sample tracking, test reports upload and instant patient access).
- Pharmacy & Inventory Management (Automated stock deduction upon dispensation, low-stock threshold alerts).
- Centralized Billing & Payment Processing (Consolidated automated invoicing, insurance co-pay discount calculation, payment gateway integration).
- Hospital Administration & MIS Reporting (Role-based access control, system audit logs, bed occupancy and financial dashboards).

5. User Classes & Stakeholders

User Class	Role Description	Key Responsibilities
Patient	Primary Customer / Healthcare Seeker	View appointment slots, book appointments, view medical history, pay bills online.
Receptionist	Front Desk Staff	Register walk-in patients, allocate appointment tokens, manage ward/bed allocation.
Doctor	Clinical Specialist	Examine patient history, record clinical diagnoses, issue e-prescriptions, order lab tests.
Pharmacist	Medical Dispensary Staff	Verify digital prescriptions, dispense medications, manage drug inventory and batch reordering.
Lab Technician	Pathology / Diagnostic Staff	Receive test orders, collect specimens, conduct diagnostic investigations, upload test results.

6. Selection and Justification of Software Process Model

For the development of the Hospital Management System (HMS), the V-Model (Verification and Validation Model) is selected as the primary lifecycle methodology, complemented by Agile Scrum practices during module-level development sprints.

Technical Justification for the V-Model:

- 1. Well-Defined and Stable Domain Requirements:** Hospital clinical workflows (patient admission, prescription generation, diagnostic testing, pharmacy dispensation, and discharge billing) follow standard institutional and medical guidelines that are thoroughly understood and frozen upfront in the IEEE 830 SRS document. The V-Model thrives in domains where requirements remain stable.
- 2. Correctness-Critical Domain:** Healthcare software directly impacts patient welfare, prescription safety, and statutory billing compliance. Defect discovery late in development carries severe liability. The V-Model enforces early verification and test planning at each stage: acceptance test planning during Requirements Analysis, system test planning during Architectural Design, integration test planning during High-Level Design, and unit test planning during Module Design.
- 3. Direct Requirement-to-Test Traceability:** Every functional requirement documented in the SRS (FR-1.0 to FR-6.0) maps one-to-one with a corresponding Verification/Validation activity, ensuring 100% test coverage and compliance auditability.
- 4. Strict Academic & Milestone Scheduling:** The project operates within a fixed semester academic timeline with defined review milestones. The linear phase-by-phase rigor of the V-Model ensures deterministic tracking and verification.
- 5. Clean Modular Architecture:** HMS naturally divides into decoupled functional modules (Registration, EHR, Pharmacy, Lab, Billing) that cleanly map to module-level unit testing (e.g., White-Box basis path testing on calculateBill).

7. Conclusion

The detailed problem statement highlights the necessity of replacing disparate manual hospital paperwork with a modern centralized Hospital Management System (HMS). By adopting the V-Model, the project ensures high reliability, early defect prevention, full requirements traceability, and rigorous verification across all clinical and financial modules.